Assignment 5: Unconsidered Metrics

Terry Benjamin Jr. | CariSurg MedTech Pathways | Week 0

Metric chosen: SpO2

The extra measure I would add is SpO2, also called oxygen saturation. It estimates the percentage of hemoglobin in the blood carrying oxygen. A pulse oximeter on the finger gives the reading. The Week 0 dataset includes FiO2, which tells us how much oxygen a patient may be receiving, but FiO2 does not tell us whether the patient's blood oxygen level is adequate. A patient can receive oxygen and still remain hypoxic.

Clinical range and caveats

For most healthy adults, oxygen saturation sits around 95% to 100%. Values below that range need context, especially for patients with chronic lung disease who may have a different target set by a clinician. A reading below 90% is a major warning sign because it suggests hypoxemia, meaning the patient may not have enough oxygen in the blood. I would not ignore that number in triage.

Triage priority

SpO2 should affect triage priority because oxygen problems can deteriorate fast. A low reading can point to pneumonia, asthma or COPD exacerbation, heart failure, pulmonary embolism, sepsis, or another breathing or circulation problem. It also helps staff decide whether the patient needs oxygen, closer monitoring, urgent review, or escalation.

How I would use it

I would record SpO2 with respiratory rate, pulse, mental status, work of breathing, and whether the reading was taken on room air or supplemental oxygen. I would also repeat the reading if the waveform is poor or the patient has cold fingers, nail polish, movement, or poor perfusion. That keeps the number useful instead of treating it as a perfect measurement.

References

MedlinePlus, Pulse Oximetry: https://medlineplus.gov/lab-tests/pulse-oximetry/

NCBI Bookshelf, Pulse Oximetry: https://www.ncbi.nlm.nih.gov/books/NBK470348/